

Muhammad Umer

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Senior Full Stack Engineer with 7+ years of experience designing and delivering scalable, high-performance web and mobile applications. Strong background in **Node.js, React, and cloud-native architectures**, with hands-on expertise in building **real-time systems, event-driven architectures, and offline-first applications**.

Proven ability to lead development teams, define system architecture, and deliver end-to-end solutions across domains including healthcare, real estate, and AI-driven platforms. Experienced in integrating complex systems, optimizing performance, and ensuring reliability in production environments.

KEY SKILLS

Architecture & System Design:

- Strong experience in **application architecture design**, defining scalable and maintainable system structures aligned with business requirements.
- Hands-on expertise in **microservices and event-driven architectures**, enabling loosely coupled and efficient systems.
- Proven ability to build **distributed systems and API ecosystems** that ensure seamless communication between services.
- Focused on designing **scalable and high-availability systems** to support growth and reliability under load.
- Experienced in developing **offline-first and real-time architectures**, ensuring functionality in low-connectivity environments with reliable data synchronization.

Cloud & Infrastructure Architecture:

- Designed and implemented **cloud architecture on Google Cloud Platform (GCP)**, focusing on scalable, secure, and high-availability systems.
- Configured **networking components including VPCs, subnets, NAT, and static external IPs**, ensuring controlled and reliable outbound and inbound connectivity.
- Managed **static regional outbound configurations**, enabling consistent IP-based communication for external integrations and APIs.
- Implemented **load balancing strategies** to efficiently distribute traffic and maintain system performance under varying loads.
- Applied **auto-scaling mechanisms** to dynamically handle traffic spikes and optimize infrastructure cost and performance.
- Worked with **Google Cloud Artifact Registry and container images**, managing build, storage, and deployment of application images.
- Evaluated and implemented **serverless (Cloud Functions), containerized (Kubernetes), and Virtual Machines** based Solutions, selecting appropriate architectures based on system requirements.
- Designed end-to-end production-grade architectures incorporating multi-tier application design (e.g., **LAMP and microservices**), managed **databases** (Cloud SQL/Firestore), **IAM-based security, monitoring & logging, Infrastructure as Code** (Terraform), and high **availability/disaster recovery strategies** aligned with Google Cloud best practices.

Back-End Development:

- Hands-on experience developing **RESTful APIs using Node.js, Express, and NestJS**, deployed via **serverless Google Firebase Cloud Functions**, focusing on scalability, maintainability, and performance.
- Skilled in **web API development with ASP.NET**, implementing robust backend solutions for enterprise applications.
- Strong expertise in **cloud platforms including GCP, AWS Lambda, and Azure**, leveraging serverless and scalable architectures.
- Implemented **authentication and authorization mechanisms** including **JWT, OAuth 2.0, and multi-factor authentication (MFA)** to secure APIs and user sessions.
- Designed **session management strategies** and secure token handling to protect user data and maintain consistent access control.
- Applied **rate limiting, throttling, and request validation** to prevent abuse, enhance security, and improve API reliability.
- Integrated **logging, monitoring, and error handling** for APIs to ensure operational stability and rapid debugging in production environments.
- Implemented **data validation and schema enforcement** using libraries like **Joi, Yup, Zod, and Express Validator**, ensuring input integrity and reducing runtime errors.
- Applied **middleware-based architecture** for handling cross-cutting concerns such as authentication, logging, validation, and error handling, improving code modularity and maintainability.

Front-End Development:

- Extensive experience developing **cross-platform mobile applications with React Native** and **dynamic web applications using React.js and Next.js**, ensuring high performance, scalability, and maintainable architecture.
- Proficient in **component-based architecture, state management (Redux, Context API), and React hooks**, delivering reusable and efficient UI code.
- Expertise in **Next.js for server-side rendering (SSR), static site generation (SSG), and API integration**, improving application performance and SEO.
- Skilled in designing **responsive and accessible interfaces** using **Ant Design (antd) and Material-UI**, following atomic design principles for consistent UI/UX across applications.
- Familiar with **Flutter development** for rapid prototyping and multi-platform deployment.
- Implemented **secure authentication flows, session management, role-based access control, and JWT/OAuth-based access** for both web and mobile applications.
- Collaborated with backend and DevOps teams to enable **CI/CD pipelines, automated testing, cloud deployment, and integration with serverless architectures** for seamless delivery.

Database Expertise:

- Extensive experience working with both **relational (SQL)** and **non-relational (NoSQL)** databases, designing optimized data models for scalability and performance.
- Hands-on expertise with **Google Cloud Firestore** and **Firebase Realtime Database**, implementing **real-time data synchronization** and offline-first capabilities.
- Proficient in **MongoDB**, including aggregation pipelines, indexing strategies, and data modelling for high-throughput applications.

- Experience with **SQL databases** (MySQL, PostgreSQL), including query optimization, stored procedures, and schema design.
- Integrated **GraphQL APIs** for efficient data querying, seamless client-server communication, and reduced over-fetching of data.
- Implemented **security and access control** at the database level, including **role-based access, JWT verification, and session management**.

AI & Intelligent Development (VIBE):

- Proficient in leveraging **AI tools and frameworks** to accelerate application development, including designing **architecture, structures, components, and reusable elements** that AI can utilize to generate additional application features.
- Experienced in using **Windsurf** and **Claude IDE** to guide AI agents for generating optimized, context-aware code and solutions.
- Applied AI-driven approaches to efficiently build **.NET Core 8 APIs**, as well as responsive **Next.js and React.js frontends**, ensuring high-performance and maintainable full-stack applications.
- Integrated multiple **intelligent agents** to produce appropriate results for backend and frontend development, improving development speed, consistency, and code quality.
- Configured and optimized a **local LLM inference stack (llama.cpp + SYCL + Intel oneAPI)** on consumer hardware, deploying a quantized **32B model** to serve a sequential multi-agent trading system for real-time market analysis and automated execution.

Version Control and DevOps:

- Used **Git for source control** with disciplined branching and pull request workflows (feature branches, reviews, and clean merge history).
- Managed **repositories in Bitbucket/Git platforms**, maintaining reliable collaboration practices (PR reviews, issue-linked commits, and release-ready mainline).
- Implemented **CI pipelines** to automatically run **linting, unit tests, E2E tests, and builds** on PRs to protect key branches.
- Supported **continuous delivery to Vercel**, promoting changes to production after successful CI validation.
- **Worked with DevOps tooling** to enforce consistent developer workflows (dependency installation via npm ci, repeatable builds, and automated checks).

Testing and Documentation:

- **Authored and maintained automated tests** using **Jest + React Testing Library** for UI components and user flows.
- **Implemented API route testing** covering **validation errors (400), success responses (200), and rate limiting (429)** scenarios.
- **Built end-to-end test coverage** with **Playwright**, including CI-friendly retries and trace capture for debugging flaky failures.
- Established **CI/CD quality gates** with **GitHub Actions**, running **lint + unit tests + E2E tests + build** on pull requests and protected branches.

Blockchain and Smart Contracts:

- Experience of developing **React-JS** applications with metamask and solarium.

- Created and deployed smart contract for NFT with **Ethereum** and **Solana** (using candy-mintmachine).

Payment Methods:

- Integrated **Stripe, Poli, Blinq** in Node.js.

Programming Languages & Tools:

- **Languages:** JavaScript, TypeScript, C, C++, C#, Java, HTML5, CSS
- **Frameworks & Libraries:** Node.js, React.js, Express.js, .NET
- **Databases & Cloud:** Firebase, Firestore, SQL, NoSQL, AWS, GCP
- **Project Management & Collaboration:** Jira, Confluence, Trello, Bitbucket, Git, GitHub, CI/CD tools (Jenkins, GitLab CI)
- **Other Tools:** Postman, Figma, Swagger, Docker

WORK EXPERIENCE

Lead Software Engineer

Aug 2023 – Feb 2026

F Three Technologies – Sharjah, UAE - Remote - www.f3technologies.eu

- Led **end-to-end software development and system architecture**, managing cross-functional teams to deliver scalable, high-quality applications within deadlines.
- Designed and developed **scalable RESTful APIs using Node.js, Express and Nest**, including modernization and migration of legacy systems into maintainable, distributed services.
- Architected **event-driven systems using Pub/Sub patterns**, enabling high-performance, loosely coupled application workflows.
- Built **responsive web and mobile applications** using React.js and React Native, ensuring seamless, high-performance user experiences across platforms.
- Engineered and managed **hybrid database architectures (Firestore, MongoDB, SQL)**, supporting real-time data synchronization, offline-first functionality, and optimized data modelling.
- Implemented **advanced offline capabilities** through synchronization strategies between NoSQL and SQL systems, ensuring reliable application performance without connectivity.
- Led **server-to-server integrations and data migrations**, maintaining data integrity and minimizing operational disruption.
- Integrated **AI-driven features**, including facial recognition for health monitoring, enhancing application intelligence and engagement.
- Developed **real-time notification systems** using Twilio and custom email services to support scalable communication workflows.
- Implemented **secure payment integrations** (Stripe, Poli, Blinq), improving transaction reliability and user experience.
- Established **CI/CD pipelines, automated testing frameworks, and streamlined deployment processes**, significantly improving code quality and release efficiency.
- Deployed and managed applications across **cloud platforms (AWS, GCP, Azure)** using Docker and Kubernetes for scalability and operational efficiency.
- Provided **technical leadership to frontend teams (React.js, Next.js & React Native)** by defining architecture, resolving complex challenges, and ensuring successful feature delivery and testing.

Senior Software Engineer

Oct 2020 – July 2023

F3 Technologies – Pakistan - Onsite - www.f3technologies.eu

- Developed and maintained **RESTful APIs using Node.js Express and Nest**, focusing on performance, scalability, and clean code practices.
- Built **responsive web and mobile applications** using React.js, Next.js and React Native, delivering consistent user experiences across devices.
- Worked with **Firestore, MongoDB, and SQL databases**, implementing efficient data models and enabling real-time data synchronization.
- Implemented **core application features including authentication, data management, and third-party integrations**.
- Contributed to **API enhancements and partial modernization of legacy systems**, improving maintainability and performance.
- Integrated **real-time features and notification systems** using services like Twilio and custom email solutions.
- Supported **payment gateway integrations** (e.g., Stripe, Poli, Blinq) to enable secure and seamless transactions.
- Assisted in **CI/CD pipeline setup and deployment processes**, improving release consistency and reducing manual effort.
- Participated in **cross-functional collaboration** with designers, QA, and backend teams to deliver features on time.
- Gained experience with **cloud platforms (GCP, AWS)** and containerization tools like Docker in development and deployment workflows.

Software Development

Jan 2020 – Oct 2020

Freelance – Remote – Pakistan

- **Blockchain Development:** Implemented smart contracts on Solana and Ethereum for buying and selling NFTs. This involved designing, deploying, and managing smart contracts to facilitate secure and efficient transactions on these platforms.
- **React.js Development:** Developed responsive websites using React.js, including integration with cryptocurrency wallets. This enables seamless user interactions with blockchain applications and supports secure wallet transactions.

Software Development

Jun 2019 – Dec 2019

Euroko International – Onsite – Pakistan

- Effectively managed IT/Network support related tasks in the organization.
- Developed an Android Application for Sales Management System.
- Designed responsive, user-friendly website in React Js.

PROJECTS

Throughout my career, I have contributed to more than 25 diverse projects. My role involved detailed work in various aspects, including project architecture, implementation, and achieving significant milestones. My experience with projects includes Real-Estate, Medical Application, Logistics, AI, Block Chain and much more. I outline the nature of my work, the architectural frameworks employed, and my key achievements for some of my project.

1. Innova Care

Innova Care is a comprehensive home care system designed to enhance the management and efficiency of home-based patient care. The system assigns tasks to nurses who visit patients at their homes. Key features of the application include:

- **Navigation Assistance:** Guides nurses with navigation within the app to ensure they reach the patient's home efficiently.
- **Attendance Tracking:** Nurses can mark their attendance using NFC tags at the patient's house or by being within a defined range (e.g., 100 meters) of the patient's location.
- **Task Management:** Nurses can create and manage notes and other details related to their assigned tasks.
- **Dynamic Feature Management:** Allows administrators to enable or disable all application features from the backend, providing flexibility to adapt the system according to operational needs.

I have successfully upgraded Innova Care from a legacy application to a modern system using the following technologies:

- **Node.js with Express:** Implemented server-side logic and API routes, enabling seamless communication between the upgraded application and backend services.
- **Google Cloud Functions:** Deployed the application on Google Cloud Functions, leveraging serverless technology to ensure high availability and scalability across multiple regions.
- **Firebase Firestore:** Utilized for real-time database management, handling data storage and synchronization effectively.
- **Offline Functionality:** Ensured the application operates offline, allowing nurses to continue their tasks without an active internet connection. Data synchronization occurs when connectivity is restored.
- **.NET and SQL Integration:** For the mobile application, migrated data from SQL to NoSQL (Firebase Firestore) using .NET API. Implemented data synchronization mechanisms to sync data back to SQL servers via API in real-time.
- **Automated Test Suites:** Set up for continuous integration and deployment (CI/CD) pipelines to ensure code quality, reliability, and early detection of issues.
- **NFC and Geofencing Technology:** Integrated NFC for attendance tracking and geofencing to verify nurse proximity to patient locations.
- **Pub/Sub Mechanism:** For efficient and scalable event-driven communication between .NET and node.js to ensure that data is sync between servers in real-time.

2. MyIndici

MyIndici is a secure, user-friendly patient portal that brings the benefits of healthcare technology directly to the patient. As a part of New Zealand's top 20 healthcare systems and a notable fintech product, MyIndici offers features like virtual consultations and more.

Key Contributions:

- **Accident/Winz Module Development:** Spearheaded the development and enhancement of the Accident/Winz module, streamlining the management of patient accident and insurance claims.
- **Data Security:** Implemented secure and efficient data handling practices for sensitive medical and insurance information, ensuring compliance with healthcare regulations.
- **UI Optimization:** Optimized the user interface, improving accessibility and overall user experience for patients.
- **Cross-Functional Collaboration:** Worked closely with cross-functional teams to ensure seamless integration and functionality within the MyIndici platform, enhancing the overall system's performance.

3. Bio Scan

Bio Scan App is an innovative health monitoring application that leverages advanced AI technology to track and analyze vital signs through facial recognition. Key features of the app include:

- **Vital Sign Monitoring:** Uses the device's camera to monitor key health indicators such as blood pressure, respiration rate, stress levels, and other vital signs through facial analysis.
- **Historical Data Storage:** Maintains a comprehensive history of the user's vital signs, allowing for trend analysis and tracking over time.
- **Report Sharing:** Enables users to generate and share detailed health reports with healthcare professionals for selected time frames, facilitating informed medical consultations.

I have developed the Bio Scan App utilizing the following technologies:

- **Node.js with Express:** Implemented server-side logic and API routes to handle communication between the app and backend services efficiently.
- **Google Cloud Functions:** Deployed the application using Google Cloud Functions, taking advantage of serverless technology to ensure high availability and scalability across multiple regions.
- **Binnah AI Integration:** Integrated **Binnah AI** for advanced facial recognition and health monitoring, providing accurate and real-time analysis of vital signs through the camera.
- **Firebase Firestore:** Utilized for real-time data storage and synchronization, ensuring that historical health data is securely stored and readily accessible.
- **Automated Test Suites:** Established continuous integration and deployment (CI/CD) pipelines with automated test suites to ensure code quality, reliability, and early detection of issues.
- **Payment Methods:** Integrated Blinq Solutions for payments.

4. Ask Cyphen

Ask Cyphen is an advanced Kibana plugin integrated with Elastic Search and ChatGPT, designed to enhance threat detection and data visualization capabilities. The plugin provides users with intuitive and actionable insights by combining powerful search capabilities with conversational AI. Key features of the plugin include:

- **Threat Visualization:** Displays threat data within Kibana dashboards, offering visual insights and analytics to help users identify and understand potential security threats.
- **AI-Powered Data Interaction:** Integrates ChatGPT with Elastic Search to enable natural language queries and interactions, making it easier for users to extract and interpret data from Elasticsearch.
- **Seamless Integration:** Combines Elasticsearch's robust search functionality with ChatGPT's conversational abilities to deliver a user-friendly interface for querying and analyzing threat data.

I played a key role in developing the Ask Cyphen plugin using the following technologies:

- **Kibana Plugin Development:** Created a custom Kibana plugin to visualize threat data, enhancing Kibana's capabilities for security and data analysis.
- **Elastic Search Integration:** Leveraged Elastic Search to index and search through threat data efficiently, providing robust search functionality within Kibana.
- **ChatGPT Integration:** Connected ChatGPT with Elastic Search to enable natural language processing, allowing users to interact with and query the threat data using conversational AI.
- **Node.js and JavaScript:** Utilized Node.js and JavaScript for developing the Kibana plugin, ensuring seamless integration and functionality within the Kibana ecosystem.

5. Eizhar

Eizhar is an advanced Property Management System designed to streamline real estate transactions and property management. It supports four key roles: landlords, tenants, dealers, and managers. Landlords can list their properties for sale or rent, complete with detailed descriptions and images. Dealers act on behalf of landlords, while tenants have the ability to search for, buy, or rent properties and can request property visits. Managers, who work in support of dealers, oversee property management tasks such as inspections and handling visit requests. Eizhar offers a comprehensive suite of features including notifications, SMS alerts, marketing emails, and chat functionality. Additionally, tenants receive reminders for unpaid rent after a specified period, ensuring timely payments and effective property management.

I have developed a robust RESTful API for the Eizhar Property Management System utilizing the following technologies:

- **Node.js with Express:** For building the server-side logic and managing API routes using Google Cloud Functions.
- **Google Firestore:** To handle real-time database needs and store property and user data.
- **Twilio:** For integrating SMS functionality and sending notifications.
- **Custom Email Solutions:** For sending personalized emails to users.
- **In-App Notifications:** To provide real-time updates and alerts within the application.
- **Background Notifications:** To ensure users receive important notifications even when the app is running in the background or not actively open.
- **Automated Test Suites:** Configured for continuous integration and deployment (CI/CD) pipelines to maintain code quality and catch issues early.
- **Pub/Sub Mechanism:** For efficient and scalable event-driven communication between different components of the system. This allows for asynchronous message delivery and helps to decouple services, improving the overall responsiveness and scalability of the application.

EDUCATION

Bs. Software Engineering

2015 – 2019

Comsats University Islamabad (CGPA 3.2)

ACADEMIC ACHIEVEMENT

Position Holder

- Position holder in 2nd and 3rd semester.

Coding Competition

- Runner-up Pre-Coder in Code-A-Than.

CERTIFICATIONS & TRAININGS**Android Development Workshop**

Workshops at Comsats University Islamabad about advance Concept of Android Development.